

Anjana Yatawara

Assistant Professor of Statistics
California State University, Bakersfield

Faculty Lead, CSUB Hub for Statistics, Applied AI, and Data Science Innovation | Hub LinkedIn

ayatawara@csub.edu | yatawara.com | ORCID | Google Scholar | LinkedIn

RESEARCH PROFILE

My research is organized around four complementary areas:

1. Time Series, Stochastic Modeling & Financial Econometrics

Statistical methodology for dependent and dynamic data, including volatility modeling, count-valued time series, and high-frequency and multi-frequency processes, with applications in finance, economics, and the sciences.

2. Statistics for Artificial Intelligence

Statistical methods for evaluating the reliability and uncertainty of AI systems, with particular emphasis on large language models, prediction, inference, and decision-making.

3. Artificial Intelligence in Mathematics & Statistics Education

Empirical and methodological research on the responsible integration of generative AI into education, with emphasis on mathematical and statistical reasoning, learning outcomes, assessment, and AI-supported tutoring.

4. Environmental Statistics & Data Science

Statistical modeling of air quality, pollutant exposure, environmental inequality, spatiotemporal processes, sensor and monitoring networks, and heterogeneous environmental data sources.

Across these areas, I work with students throughout the research process, from data analysis and statistical computing to scientific writing and publication. I also collaborate with researchers across disciplines on both methodological and applied statistical problems.

ACADEMIC APPOINTMENT AND EDUCATION

Assistant Professor of Statistics

California State University, Bakersfield, Bakersfield, CA

2023–present

Ph.D., Mathematics (Statistics Emphasis)

Missouri University of Science and Technology, Rolla, Missouri

Research emphasis: modeling time series with conditional heteroscedastic structure

Advisor: Dr. V. A. Samaranayake

2023

B.Sc. Special Degree, Statistics

University of Peradeniya, Sri Lanka

Second Class Honors (Upper Division)

2017

ACADEMIC LEADERSHIP AND PROGRAM DEVELOPMENT

Founder & Faculty Lead, CSUB Hub for Statistics, Applied AI & Data Science Innovation

2025–present

- Founded and lead an interdisciplinary Hub supporting statistics, applied AI, data science, student research, research computing, and cross-disciplinary collaboration.
- Established a physical home for the Hub and developed computing infrastructure supporting undergraduate and faculty research.

Founder & Organizer, Statistics, Applied AI & Data Science Colloquium Series

2025–present

- Founded and organize an interdisciplinary colloquium series bringing external researchers and professionals to CSUB to engage students and faculty in statistics, data science, AI, scientific computing, and applied research.
- Organized the first three colloquia with Thomas A. DeFanti on CENIC AIR and the National Research Platform (Nov. 2025), Bhash Abeysinghe of the American Institutes for Research on AI, NLP, and agentic systems (Apr. 2026), and two NASA Ozone Where We Live research scientists on environmental data science (Sept. 2026).

Statistics Coordinator, Department of Mathematics**2026–present**

- Coordinate MATH 1209: Statistics in the Modern World and MATH 2200: Introduction to Statistical Concepts and Methods across multiple sections, including course expectations, content coverage, shared resources, instructor support, and communication.
- Developed a shared instructional system for MATH 2200 and MATH 1209 with course materials, R resources, assignments, assessments, classroom activities, projects, and browser-based statistical computing through CSUB JupyterHub.

Co-Developer & Instructor, MATH 4230: Applied Statistical Methods for Data Science**2025-2026**

- Co-developed MATH 4230 with Statistics faculty as an upper-division applied statistics and data science course and taught the course in Spring 2026.
- Developed the Spring 2026 implementation, including the syllabus, instructional materials, computational labs, assessments, and an open-ended capstone centered on real data, reproducible workflows, and statistical communication. Used R, Python, JupyterHub, and structured approaches to responsible AI use. Participated in the 2026 National Workshop on Data Science Education at UC Berkeley to benchmark the course and broader curriculum.

MATH 3218: The Science of Data, Statistics, and Artificial Intelligence**2025**

- Participated in faculty review and refinement of the new upper-division GE course led by Dr. Eduardo Montoya and served as the named proposer responsible for entering and advancing the proposal through CourseLeaf

Statistical Data Science Concentration and Proposed B.S. in Statistical Data Science**2025-present**

- Work with Statistics and Mathematics faculty on CSUB's Statistical Data Science curriculum, including the existing concentration and the current effort to develop a standalone B.S. in Statistical Data Science.
- Contributed to degree requirements, prerequisite structure, elective pathways, comparisons with programs at other universities, student-demand survey analysis, and proposal development.

Proposed General Education Course: Inside AI – The Mathematics of Large Language Models**Initial proposal**

- Developed an initial proposal and course material for a lower-division Area 2: Mathematical Concepts & Quantitative Reasoning course that teaches mathematics and statistics through the inner workings of modern language models.
- This course on tokens, embeddings, vectors and matrices, attention, probability, sampling, uncertainty, numerical precision, quantization, model size, and statistical evaluation. The concept adapts hands-on small-model teaching ideas encountered through Eric Van Dusen's UC Berkeley/NWDSE materials to a mathematics-and-statistics GE setting.

Founder & Organizer, CSUB AI & Data Science Research Group**2026–present**

- Established a cross-departmental faculty research group connecting statistics, computer science, artificial intelligence, institutional research, and related areas; organize research meetings and support collaborative projects, student involvement, and grant opportunities.

Research Computing & AI Infrastructure Development**2025–present**

- Contributed to research-computing infrastructure supporting statistics, data science, and AI research at CSUB, including connection to the National Research Platform/Nautilus ecosystem for GPU and high-performance computing.
- Participated in CSUB's AWS Cloud & AI initiative, contributing statistical evaluation and benchmarking, curriculum-integration ideas, and student-training plans to a project supported by substantial AWS cloud-computing credits.

AI and Data Science Workshops and Training**2025–present**

- Developed and delivered a sustained program of workshops, seminars, and training activities on responsible and useful applications of generative AI in teaching and research.
- Selected activities include *Teaching With AI: Helping Students Learn Statistics With Custom GPTs and Prompts*, *AI as a Thinking Partner in Statistics*, and *Effective Ways to Introduce AI in Your Classroom at CSUB*.
- Developed course-specific AI tutoring approaches to support mathematical and statistical reasoning and contributed to broader campus AI activities, including OpenAI University Day at CSUB.

SERVICE**PROFESSIONAL**

- **Member, ADSA AI Tutoring Study Working Group**, Alliance for Data Science and AI — multi-institution working group examining AI tutoring and AI-supported learning in statistics and data science education. *2026–present*.
- **Mentor, ASA Section on Statistics and Data Science Education (SSDSE) Mentoring Program**, American Statistical Association. *2026–2027*.
- **Program Committee Member and Reviewer**, Artificial Intelligence × Humanities, Education, and Art (AIxHEART) 2026. *2026*.

- **Reviewer**, IEEE International Conference on Tools with Artificial Intelligence (ICTAI). 2026.

UNIVERSITY

- **Academic Affairs Committee (AAC)**, Academic Senate — appointed member of the university standing committee concerned with academic policy, programs, curricula, and related academic matters. 2026–2027.
- **General Education Curriculum Committee (GECCo)** — elected by acclamation as a Mathematics/NSME representative to complete an existing term. *May 2026–May 2027*.

DEPARTMENT

- **Founding Faculty Advisor, CSUB Data Science & Applied AI Society** — recruited founding student officers and guided development and registration of the new student organization. 2026–*present*.
- **Mathematics Department Website Subcommittee** — member; coordinated collection of faculty and staff information and distributed the departmental faculty/staff webpage survey. 2024–2025.

OUTREACH AND COMMUNITY ENGAGEMENT

- **Co-Organizer, Data Science Exploration Challenge, Lee Webb Math Field Day** — developed a new high-school data science competition, including the dataset, data dictionary, participant materials, judging framework, school communication, and event implementation. *March 2026*.
- **Volunteer, Kern County Chapter MATHCOUNTS Competition** — supported competition operations and student activities, including scoring support. 2025, 2026.
- **Volunteer, Future 'Runner Day / Meet Your Major** — represented Mathematics in prospective-student and family outreach. *March 2026*.
- **Volunteer, NSME Open House** — represented the Mathematics Department to prospective students and families. *October 2025*.
- **Faculty Volunteer, Math Club Runner Night**. *March 2025*.

TEACHING EXPERIENCE

Instructor of record for **17 sections across six distinct courses** at California State University, Bakersfield over seven consecutive semesters (Fall 2023–Fall 2026), spanning general-education statistics, probability and mathematical statistics, statistical computing, regression, and upper-division data science. Previously taught engineering statistics and engineering calculus at Missouri University of Science and Technology (2017–2023).

California State University, Bakersfield — Instructor of Record 2023–present

Course	Title	Level	Terms	Sec.
MATH 2200	Introduction to Statistical Concepts and Methods	Lower-division, GE service	F23–F26, every term	7
MATH 3200	Probability Theory	Upper-division core	S24, F25, S26, F26	5
MATH 3210	Applied Statistical Computing	Upper-division; graduate cross-list (MATH 5210)	F23, F24	2
MATH 4200	Mathematical Statistics	Upper-division core	S25	1
MATH 4210	Regression Modeling and Analysis	Upper-division elective	F26	1
MATH 4230	Applied Statistical Methods for Data Science	Upper-division elective; course co-developed by instructor	S26	1

Missouri University of Science and Technology — Graduate Teaching Assistant / Instructor 2017–2023

- **STAT 3113: Applied Engineering Statistics** — lecture sections, typically two per term; full responsibility for lectures, assessment, and grading.
- **MATH 1215: Calculus for Engineers II** — lecture sections; full responsibility for lectures, assessment, and grading.
- **MATH 1214: Calculus for Engineers I** — shared instructional responsibility.

University of Peradeniya, Sri Lanka — Instructor and Tutor

- **Instructor:** CS 100 Computer Applications; ST 103 Statistics Applications I.
- **Tutor:** ST 301 Regression Analysis; ST 305 Multivariate Methods I; ST 403 Statistics for Bioinformatics.

STUDENT MENTORING AND RESEARCH SUPERVISION

UNDERGRADUATE RESEARCH SCHOLAR PROGRAMS

Faculty Research Mentor, CV Pathway – Summer Research Apprenticeship for Doctoral Programs 2024, 2025

- Mentored **Tom Regpala** in 2024 and **Emily DeJesus** and **Noah Gallego** in 2025 through intensive summer research and doctoral-preparation experiences.
- Supervised research design, literature review, data acquisition, statistical and computational analysis, research writing, and poster/presentation development.
- Continued several collaborations beyond the summer program into sustained undergraduate research and publication activity.

Faculty Research Mentor, Chevron Summer Undergraduate Research Experience (SURE) Summer 2025

A Customizable AI Pipeline for Academic Excellence: Locally Hosted Language Models in Higher Education

- Mentored undergraduate researchers in the development and evaluation of locally hosted large language models for higher education, including retrieval-augmented generation (RAG), model fine-tuning, quantization, GPU-based deployment, and comparison with commercial cloud-based AI systems.

Principal Investigator & Undergraduate Research Mentor, CES Mini-Grant 2025–2026

Air Pollution Disparities in California, 2000–2025: Examining Environmental Inequities Across Income, Poverty, and Minority Populations

- Supervised undergraduate research in environmental statistics, air-pollution exposure inequality, statistical computing, and large-scale demographic/environmental data integration.
- Student research contributed to multiple peer-reviewed publications in environmental statistics and environmental health.

INDIVIDUAL UNDERGRADUATE RESEARCH SUPERVISION

Tom Regpala 2024–2026

Environmental statistics, air-pollution exposure inequality, statistical computing, and environmental data integration. Began through CV Path 2024 and continued as a sustained research collaboration. Student co-first author on peer-reviewed work published in *Atmospheric Environment: X*; continued mentoring included research writing and doctoral-study preparation.

Eric Palafox 2025–2026

Environmental statistics and computational data analysis. Supervised large-scale monitor–satellite air-quality analysis, R programming, statistical methodology, and reproducible workflows. Student co-first author on peer-reviewed work published in *Atmospheric Environment: X*.

Kayla Ko 2025–2026

Environmental inequality and statistical data science. Supervised research on income-based pollution-exposure disparities across California counties. Student first author on peer-reviewed work published in *Environmental Research: Health*.

Christian Rodriguez 2025–2026

Environmental data science and applied AI; supported through Chevron SURE and ELEVATE. Student coauthor on peer-reviewed work published in *Environmental Research: Health* and co-presenter of undergraduate research on open-source/local large language models.

Noah Gallego 2025–2026

Machine learning, time-series forecasting, and computational data science; CV Path 2025 Research Scholar. Supervised hybrid GRU-CNN influenza forecasting, computational implementation, statistical evaluation, and manuscript development.

Emily DeJesus 2025–2026

Environmental data science; CV Path 2025 Research Scholar. Supervised literature review, environmental/demographic data analysis, interpretation, research writing, and poster preparation; continued graduate-school and research-pathway mentoring.

Juan Rodriguez Aguilar 2024–2026

Applied AI, forecasting, and open-source/local large language models. Supervised research development, computational experimentation, poster preparation, and communication of findings. Co-presented undergraduate research through the CSUB Mathematics Seminar.

Chase Davis & Brian Martinez Fall 2026–present

Undergraduate mathematics research in cryptography and LLM text watermarking; project currently in the initial research-development stage.

STUDENT RESEARCH PRESENTATIONS AND COMPETITIVE ACHIEVEMENTS

CSU Channel Islands Plot-A-Thon February 2026

- Mentored and helped organize participation of eight CSUB students, including recruitment, team formation, funding, travel, and lodging coordination.

- CSUB Team 1 – **Juancarlos Sandoval, Krrithik Ezhilarasan, Christine Leanna Bonoan, and Sean Toledo** – won **Best Overall, Best in Data Analysis, and Best in Data Communication**.

CSUB Mathematics Seminar – Undergraduate AI Research

November 19, 2025

- Mentored **Tom Regpala, Juan Rodriguez Aguilar, and Christian Rodriguez** in presenting undergraduate AI and data-science research; Aguilar and Rodriguez presented *CSUB-GPT: Open-Source Local LLMs for Campus Use*.

Joint Statistical Meetings

2025

- Undergraduate research with **Noah Gallego** contributed to presented work on hybrid deep-learning methods for forecasting influenza-like illnesses.

CV Path Research Symposia

2024–2025

- Supervised preparation of undergraduate research reports, posters, and presentations associated with the CV Path research program.

GRADUATE AND PROFESSIONAL PATHWAY MENTORING

CSU Chancellor’s Doctoral Incentive Program (CDIP)

2025–2026

Faculty mentor and recommender for **Tom Regpala’s** 2026–27 CDIP doctoral-pathway application.

Cal-Bridge Graduate Pathway Mentoring

2025

Supported **Emily DeJesus** in pursuing Cal-Bridge and communicated with program faculty regarding her research preparation and graduate-school pathway.

Graduate School and Career Mentoring

Ongoing

Mentor undergraduate researchers in graduate and Ph.D. applications, research CVs, statements of purpose, fellowships, internships, employment applications, and letters of recommendation.

Regional Research Collaboration

Local, Task-Specific Language Model for Automated Essay Grading

2026–present

Collaboration with Dr. Jonathan P. Brown, Professor of Mathematics, Bakersfield College, Bakersfield, CA

- Developing a resource-efficient local language model for an existing automated essay-grading application, with shared computing resources and participation of Bakersfield College student researchers.
- Investigating neural-network, small language model, and model-distillation approaches for task-specific essay grading and evaluation.

GRANTS, FELLOWSHIPS & SPONSORED PROJECT ACTIVITY

FUNDED AWARDS

California Education Learning Lab – AI FAST Challenge

2025–2026

Co-Principal Investigator, *ELEVATE: Enhancing Learning Experiences Via AI Techniques*. **\$150,000**. Interdisciplinary project integrating generative AI and structured AI tutoring into mathematics, statistics, and other university courses; included student research, faculty development, and evaluation of AI-supported learning.

CSUB Environmental Studies – CES Mini-Grant

2025–2026

Principal Investigator, *Air Pollution Disparities in California, 2000–2025: Examining Environmental Inequities Across Income, Poverty, and Minority Populations*. Funded environmental statistics project supporting undergraduate research, large-scale air-quality and demographic data analysis, and peer-reviewed scholarship.

CSUB Center for Entrepreneurship & Innovation – Faculty R&D Fellowship

2026–2027

Faculty R&D Fellow / Project Lead, *KernelStats: AI-Powered Statistical Analysis Platform*. **\$7,000**. Competitive R&D fellowship supporting development, testing, intellectual-property planning, and commercialization of an AI-assisted statistical analysis platform.

Amazon Web Services – Cloud & AI Research Credit Award

2026

Faculty Collaborator / Project Team Member. **\$200,000 in AWS cloud credits awarded to the collaborative CSUB project**. Contributed statistical evaluation and benchmarking expertise, curricular-integration planning, and student-research applications.

Chevron Summer Undergraduate Research Experience (SURE)

Summer 2025

Faculty Research Mentor, *A Customizable AI Pipeline for Academic Excellence: Locally Hosted Language Models in Higher Education*. **\$7,000 faculty mentor award**. Supervised undergraduate research on locally hosted large language models, retrieval-augmented generation (RAG), model fine-tuning, quantization, GPU-based deployment, and development of institution-specific AI applications.

SUBMITTED / UNDER REVIEW

U.S. Department of Education – Title III Strengthening Institutions Program (SIP) 2026

CATALYST: Cultivating Adaptive Teaching And Learning with AI-Supported Technologies. AI Learning Companion Tool Developer (0.33 FTE); Key Personnel. Revised institutional proposal submitted June 2026; decision pending. Responsible for developing and maintaining the Adaptive AI Learning Companion, a scaffolded AI tutoring system for STEM gateway courses, building on prior mathematics and statistics AI-tutoring research and the ELEVATE project.

U.S. Department of Education — Supporting Effective Educator Development (SEED) 2026

Proposal Team Contributor; Cruz, A. C. (PI), Yatawara, A., et al. FY 2026 SEED Competition, Assistance Listing 84.423A. Contributed quantitative research design, assessment, mathematics/statistics learning materials, analysis of implementation and participant outcomes, and proposal/budget development.

PROPOSALS IN PREPARATION

National Science Foundation – Mathematical Foundations of Artificial Intelligence (MFAI), NSF 24-569 2026

Co-Principal Investigator; PI: Jeremy Woods. Cross-CSU interdisciplinary proposal investigating mathematical and statistical foundations of artificial intelligence, including reliability, robustness, uncertainty, interpretability, computational efficiency, and responsible AI. Planned submission: **October 2026.**

UNFUNDED PROPOSALS

California State University – Learning Innovation for Future-Forward Teaching (LIFT), Tier 2 2026

Principal Investigator, A CSUB Interactive Learning Platform for MATH 2200. \$20,000 requested. Proposed an integrated, open, and interactive statistics-learning environment combining a digital coursebook, computational tools, assessments, browser-based statistical computing, and a course-material-grounded ChatGPT tutor designed to provide students with context-specific support using the instructor's own MATH 2200 instructional materials. **Not selected for funding.**

CSUB SPARKS – Students Partnering with AI for Regional Knowledge and Service 2026

Principal Investigator / Faculty Applicant, MATH 3209: Statistical Measures of Inequality in Society. Proposed an AI-integrated, project-based redesign of the upper-division GE course. **Not funded because MATH 3209 was not scheduled for Fall 2026, a program eligibility requirement.**

NSME Seed Research & Writing Grant 2025

Principal Investigator, Enhancing Volatility Modeling: High-Frequency Data Analysis and Theoretical Development of FIGJRGARCH Models. Proposed theoretical and empirical development of new volatility models using high-frequency financial data. **Not selected for funding.**

U.S. Department of Education – FIPSE Special Projects Program 2025

Co-Principal Investigator, CATALYST: Cultivating Adaptive Teaching And Learning with AI-Supported Technologies. Institutional proposal to expand responsible AI-supported teaching and learning across STEM disciplines. Led development of mathematics/statistics and scaffolded AI-tutoring components, including the SMARTS framework and course-integrated AI learning companion. **Submitted December 2025; not funded.**

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

1. **Yatawara, A.** (2026). Good portfolios from bad forecasts: The anatomy of LLM volatility estimates. *Finance Research Letters*, 109, 110572. doi:10.1016/j.frl.2026.110572
2. **Yatawara, A.** (2026). Does trading volume improve long-term volatility forecasts? Evidence from the MF2-GARCH framework. *Journal of Forecasting*. doi:10.1002/for.70199
3. Regpala, T., Palafox, E., & **Yatawara, A.** (2026). Air pollution exposure inequality across U.S. income and racial groups, 2000–2023: A hybrid monitor–satellite analysis with two new shape diagnostics. *Atmospheric Environment: X*, 31, 100492. doi:10.1016/j.aeaoa.2026.100492
4. Ko, K., Rodriguez, C., & **Yatawara, A.** (2026). Income-based exposure disparities across California counties, 2000 to 2023: A generalizable statistical framework. *Environmental Research: Health*, 4(1), 011002. doi:10.1088/2752-5309/ae4134
5. Abhayaratne, V., Hao, W., Ye, C., **Yatawara, A.**, Hopke, P. K., Li, J., & Wang, Y. (2026). Evaluating fine-scale air-quality heterogeneity using a low-cost multipollutant sensor network in Twin Cities, Minnesota. *ACS ES&T Air*, 3(4), 1057–1068. doi:10.1021/acsestair.5c00439

ACCEPTED FOR PUBLICATION

6. Cruz, A. C., **Yatawara, A.**, Mishra, M., & Wang, J. J. (2026). Structured AI-Tutoring for Computer Architecture Courses. *ASEE Computers in Education Journal*, 2026 FIE Special Issue. Accepted for publication, June 23, 2026.

CONFERENCE PROCEEDINGS AND PEER-REVIEWED CONFERENCE PAPERS

7. Cruz, A. C., **Yatawara, A.**, Mishra, M., & Wang, J. J. (2025). WIP: Structured AI Tutoring in Engineering Education. *Proceedings of the 2025 IEEE Frontiers in Education Conference (FIE)*, 1–5. doi:10.1109/FIE63693.2025.11328293
8. Cruz, A. C., **Yatawara, A.**, Mishra, M., & Wang, J. J. (2025). Computer-Aided Instruction for K–12 Teachers: A Cognitive Apprenticeship Approach to LLM Integration. *2025 Artificial Intelligence x Humanities, Education, and Art (AIxHEART)*, 13–16. doi:10.1109/AIxHEART65685.2025.00009
9. **Yatawara, A.**, & Samaranayake, V. A. (2022). The Multiplicative Factor Multi-Frequency Exponential GARCH ((MF)2-EGARCH). *Proceedings of the Joint Statistical Meetings, Business and Economic Statistics Section*.
10. **Yatawara, A.**, & Samaranayake, V. A. (2021). The Asymmetric Hyperbolic Generalized Autoregressive Conditional Heteroscedastic (A-HYGARCH) Model. *Proceedings of the Joint Statistical Meetings, Business and Economic Statistics Section*.

MANUSCRIPTS UNDER REVIEW, REVISION, AND EDITORIAL CONSIDERATION

ARTIFICIAL INTELLIGENCE IN MATHEMATICS & STATISTICS EDUCATION

1. **Yatawara, A.**, Wang, J., Cruz, A. C., & Mishra, M. (2026). From Answer Generators to Thinking Partners: Scaffolded LLM Tutors in Statistics Education. *Journal of Statistics and Data Science Education*. Third revision submitted.
2. **Yatawara, A.**, Wang, J., Cruz, A. C., & Mishra, M. (2026). From an Apparent Gain to a Composition Artefact: A Three-Condition Study of Generative AI in an Undergraduate Proof-Based Mathematics Course. *Computers & Education: Artificial Intelligence*. Submitted; under editorial assessment.

STATISTICS FOR ARTIFICIAL INTELLIGENCE

1. **Yatawara, A.** (2026). Empirical Asset Pricing via Large Language Models. *The Journal of Finance and Data Science*. Revised manuscript submitted.

TIME SERIES, STOCHASTIC MODELING & FINANCIAL ECONOMETRICS

1. **Yatawara, A.** (2026). When Variance Arrives: Intraday Timing Information in Realized-Variance Forecasting. *Journal of Econometrics*. Under review.
2. **Yatawara, A.** (2026). The Timescale Structure of Volatility Memory. *Journal of Financial Econometrics*. Under consideration.
3. **Yatawara, A.** (2026). The Shape of Volatility Memory. *Quantitative Finance*. Under consideration.
4. **Yatawara, A.** (2026). A Multiplicative-Factor Multi-Frequency Conditional-Mean Model for Count Time Series. *Journal of Time Series Analysis*. Revised manuscript submitted.
5. **Yatawara, A.** (2026). Quasi-Maximum Likelihood Estimation of EGARCH(p,q): Continuous Invertibility, Consistency, and Asymptotic Normality under Verifiable Conditions. *Econometric Theory*. Under editorial assessment.
6. **Yatawara, A.** (2026). Do Volatility Components Explain the Heavy Tails of GARCH Residuals? *International Review of Economics & Finance*. Under consideration.

SPATIAL & ENVIRONMENTAL STATISTICAL METHODOLOGY

1. **Yatawara, A.** (2026). Lagged Causal Effects from Spatially Aggregated Data: A Change-of-Support Distributed-Lag Framework with Design Diagnostics. *Spatial Statistics*. Under consideration.

SELECTED WORKS IN PROGRESS

1. **Yatawara, A.** Prediction-Powered Inference with Model-Assisted Validation Labels.
2. **Yatawara, A.**, & Samaranayake, V. A. (MF)2-GARCH-A: Volatility Modeling with a Sign-Sensitive Long-Run Component.
3. **Yatawara, A.**, & Samaranayake, V. A. Multiple-Regime Hyperbolic GARCH (MR-HYGARCH).
4. **Yatawara, A.**, Gallego, N., & Ratnayake, I. Improving Influenza Forecasting: A GRU-CNN Hybrid Model with Yeo-Johnson Scaling.

CONFERENCE PRESENTATIONS

PEER-REVIEWED AND CONTRIBUTED PRESENTATIONS

- **2026** — **Yatawara, A.**, Wang, J., Cruz, A. C., & Mishra, M. “From Answer Generators to Thinking Partners: Scaffolded LLM Tutors in Statistics Education.” Joint Statistical Meetings, Boston, MA, August 6, 2026. Session: *AI in the Classroom: Research and Exploration*. Presenter: Anjana Yatawara.
- **2026** — Cruz, A. C., **Yatawara, A.**, Mishra, M., & Wang, J. J. “Prompt-Engineered Cognitive Apprenticeship for In-Service K–9 Teacher Learning.” Annual Meeting of the American Educational Research Association (AERA), Los Angeles, CA. Roundtable presentation.
- **2025** — LeBlanc-Grappendorf, S., Conlon, J., Johnson, M., **Yatawara, A.**, Wang, J. J., Cruz, A. C., & Mishra, M. “Scaffolded Socratic Tutoring with a Retrieval-Augmented Mistral LLM for Psychology Laboratory Courses.” IEEE International Conference on Artificial Intelligence x Humanities, Education, and Art (AIxHEART), Laguna Hills, CA.
- **2025** — Cruz, A. C., **Yatawara, A.**, Mishra, M., & Wang, J. J. “Computer-Aided Instruction for K–12 Teachers.” IEEE International Conference on Artificial Intelligence x Humanities, Education, and Art (AIxHEART), Laguna Hills, CA.
- **2025** — Cruz, A. C., **Yatawara, A.**, Mishra, M., & Wang, J. J. “Work-in-Progress: Structured AI Tutoring in Engineering Education.” ASEE/IEEE Frontiers in Education Conference, Nashville, TN.
- **2025** — **Yatawara, A.** “Hybrid LSTM Deep Learning Model for Forecasting Influenza-Like Illnesses.” Contributed Paper Presentation, Infectious Disease Epidemiology Section, Joint Statistical Meetings, Nashville, TN.
- **2022** — **Yatawara, A.**, & Samaranayake, V. A. “The Multiplicative Factor Multi-Frequency Exponential GARCH ((MF)2-EGARCH).” Contributed Paper Presentation, Business and Economic Statistics Section, Joint Statistical Meetings, Washington, DC.
- **2021** — **Yatawara, A.**, & Samaranayake, V. A. “The Asymmetric Hyperbolic Generalized Autoregressive Conditional Heteroscedastic (A-HYGARCH) Model.” Contributed Speed Presentation, Time Series and Finance Session, Joint Statistical Meetings, Virtual Conference.

INVITED PRESENTATIONS, SEMINARS & WORKSHOPS

ARTIFICIAL INTELLIGENCE & STATISTICS EDUCATION

- **2025** — “ELEVATE: Enhancing Learning Experiences Via AI Techniques—Findings from CSUB.” Invited presentation/panel, AI in Education Day, National Workshop on Data Science Education, University of California, Berkeley, June 24, 2025.
- **2025** — “Build Your AI Teaching Assistant.” Session Lead, NextTech Kern, California State University, Bakersfield, October 2, 2025.
- **2025** — “AI in Business.” Session Lead, NextTech Kern, California State University, Bakersfield, October 2, 2025. In collaboration with Keelan Schule, Solutions Engineer, OpenAI.
- **2025** — “Teaching With AI: Helping Students Learn Statistics With Custom GPTs and Prompts.” Mathematics Seminar, California State University, Bakersfield, April 2, 2025.
- **2025** — “Hands-On AI Workshop: Custom GPTs and Effective Prompt Engineering for Educators.” Invited Talk, NSME Teacher-Scholar Lecture Series, California State University, Bakersfield, April 11, 2025.
- **2025** — “AI Workshop for First-Year Seminar Students and Course Coordinators.” Faculty training workshop, California State University, Bakersfield, October 13, 2025.
- **2025** — “AI as a Thinking Partner in Statistics: Results from Spring and Fall 2025.” Mathematics Seminar, California State University, Bakersfield, November 5, 2025.
- **2025** — “Effective Ways to Introduce AI in Your Classroom at CSUB.” NSME Teacher-Scholar Lecture Series, California State University, Bakersfield, March 6, 2025. With Alberto Cruz and Maruti Mishra.
- **2025** — “Teaching and Learning in the AI Era.” NSME Teacher-Scholar Summer Workshop, California State University, Bakersfield, August 19, 2025. With Alberto Cruz.
- **2025** — “Accelerate Your Intelligence.” Panelist, California State University, Bakersfield, November 20, 2025.

DISSERTATION

Yatawara, A. (2023). *The Multiplicative Factor Multi-Frequency Exponential GARCH ((MF)2-EGARCH)*. Ph.D. dissertation, Missouri University of Science and Technology.